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Medical application: Renal work

Renal work is the **energy expended** by the kidneys to filter the blood and to secrete or reabsorb solutes such as sodium, urea, or glucose.

To transfer n_0 osmoles of a substance from the blood (pl) to the urine (u), this work is given by:

$$\Delta W = n_o RT \ln\left(\frac{\omega_u}{\omega_{pl}}\right)$$

ω_u and ω_{pl} :osmolarities of the solute in the urine and plasma, respectively.

Exercise 8:

A patient with moderate renal insufficiency eliminates a fixed amount of $0,36mol$ urea per 24 hours.

1. The patient produces a urine volume of $0,6l/j$, and his uremia (the mass concentration of urea in the blood) is $0.6g/l$.

a – Calculate the urinary osmolarity of urea (ω_u).

b – Determine the minimum power exerted by each kidney to eliminate this urea.

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2. The patient is asked to drink water so that his urine volume increases to $1,8l/j$. It is assumed that the amount of urea to be eliminated and the uremia remain unchanged.

a – Calculate the new urinary osmolarity of urea (ω'_u).

b – Determine the new power provided by each kidney.

3. By comparing the two power values, explain why drinking plenty of of water protects this patient's kidneys.

Given: $M_{urea} = 60g/mol$

Solutions:

1. $a. C_u = \frac{n_0}{V} = \frac{0.36}{0.6} = 0.6mol/l \Rightarrow \omega_u = 0.6osmol/l$

$b.\ C = \frac{C_p}{M} \Rightarrow C_{pl} = 0.01mol/l \Rightarrow \omega_{pl} = 0.01osmol/l \Rightarrow \frac{\omega_u}{\omega_{pl}} = 60$

$$P = \frac{1}{2}\frac{\Delta W}{\Delta t} = \frac{1}{2}\frac{RTn_o \ln(\omega_u/\omega_{pl})}{\Delta t} = \frac{8.31\times310\times0.36\times\ln(60)}{172800} = 21.97mW$$

2. $a.\ \omega'_u = \frac{n_0}{V'} = \frac{0.36}{1.8} = 0.2osmol/l \Rightarrow \frac{\omega'_u}{\omega_{pl}} = 20$

$$b.\ P' = \frac{1}{2}\frac{RTn_o \ln(\omega'_u/\omega_{pl})}{\Delta t} = \frac{8.31\times310\times0.36\times\ln(20)}{172800} = 16.08mW$$

3. $P' < P \Rightarrow W' < W$

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Drinking water increases urine volume and decreases the osmolarity of urea in the urine; consequently, the work imposed on the nephrons is significantly reduced. This is crucial to avoid exhausting the remaining nephrons in a patient with renal insufficiency.

H- TRANSPORT PHENOMENA

The **exchange** of water and solutes across membranes, between the intracellular and extracellular compartments of the body, are **transport phenomena**. They depend on:

- the **nature** of the transported particles (size, charge, shape).
- the **osmolarity** of the solutions.
- the **pressures** exerted on either side of the membrane.

1 - TYPES OF TRANSPORT

3 types of transports:

- **passive**, via thermal agitation of the particles due to:

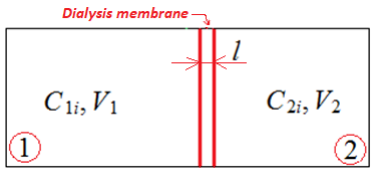
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- a difference in concentrations ⇒ **diffusion**.
- a difference in pressures ⇒ **convection**.
- a difference in electrical potentiels ⇒ **migration**.
- **Facilitated passive transport** by proteins (**membrane transporters**) of certain particles through specific channels, **without energy expenditure**.
- **active transport** of particles by proteins (membrane transporters) consuming chemical energy (ATP hydrolysis).

2 - PASSIVE TRANSPORTS

2-1 MOLAR FLOW RATE OF THE SOLUTE

Same solute in (1) and (2) with initial concentrations respectively C_{1i} and C_{2i}.



Initial state (i)

$C_{1i} > C_{2i}$

$n_{1i} = C_{1i}V_1 \qquad n_{2i} = C_{2i}V_2$

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By convention, it is assumed that solute flows occur first, followed by liquid flows.

$C_{1i} > C_{2i} \Rightarrow$ solute **diffuses** from $(1) \rightarrow (2)$ until the concentrations are **equal**.

final state (f)

$$C_{1f} = C_{2f} \equiv C_\infty$$

$$n_{1f} = C_{1f}V_1$$

$$n_{2f} = C_{2f}V_2$$

The total number of moles of the solute does not change:

$$\Rightarrow n_{1i} + n_{2i} = n_{1f} + n_{2f}$$

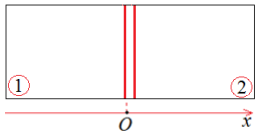
$$C_{1i}V_1 + C_{2i}V_2 = C_{1f}V_1 + C_{2f}V_2$$

$$= C_\infty(V_1 + V_2) \Rightarrow C_\infty = \frac{C_{1i}V_1 + C_{2i}V_2}{V_1 + V_2}$$

$$C_\infty = \frac{C_{1i} + C_{2i}}{2} \quad if \quad V_1 = V_2$$

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We will restrict our analysis to unidirectional flows along the x-axis, with its origin O at the membrane.



- The **molar flow** rate of the solute is defined as:

$$Q = \frac{dn}{dt} = -DS'\frac{dC}{dx} \text{ : } \textbf{\textit{Fick}'s law.} \qquad [Q] = mol/s$$

D : **diffusion** coefficient of the solute. $[D] = m^2/s$

$\frac{dC}{dx}$: concentration gradient. $\left[\frac{dC}{dx}\right] = mol/m^4.$

S' : permeable pore surface area.

- The **initial molar flow** rate is written:

$$Q_i = -DS'\frac{dC_i}{dx}$$

$$\left.\begin{array}{l} dC_i \equiv \Delta C_i = C_{2i} - C_{1i} < 0 \\ dx \equiv l \end{array}\right\} \Rightarrow Q_i = -DS'\frac{\Delta C_i}{l} \equiv \frac{DS'}{l}|\Delta C_i|$$

Remark

Fick's law can also be written in terms of mass concentration:

$$Q = \frac{dm}{dt} = -DS' \frac{dC_p}{dx} \qquad [Q] = kg/s.$$

$\frac{dm}{dt}$: mass flow rate.

2-2 DIFFUSIF FLUX OF SOLUTE

$$J_d = \frac{Q}{S} = -D\frac{S'}{S}\frac{dC}{dx} \qquad [J_d] = mol/s.m^2$$

J_d : molar flow rate per unit area.

S : membrane surface.

Memo

Rate of $X = \frac{X}{time}$

Flux of $X = \frac{X}{Surface}$